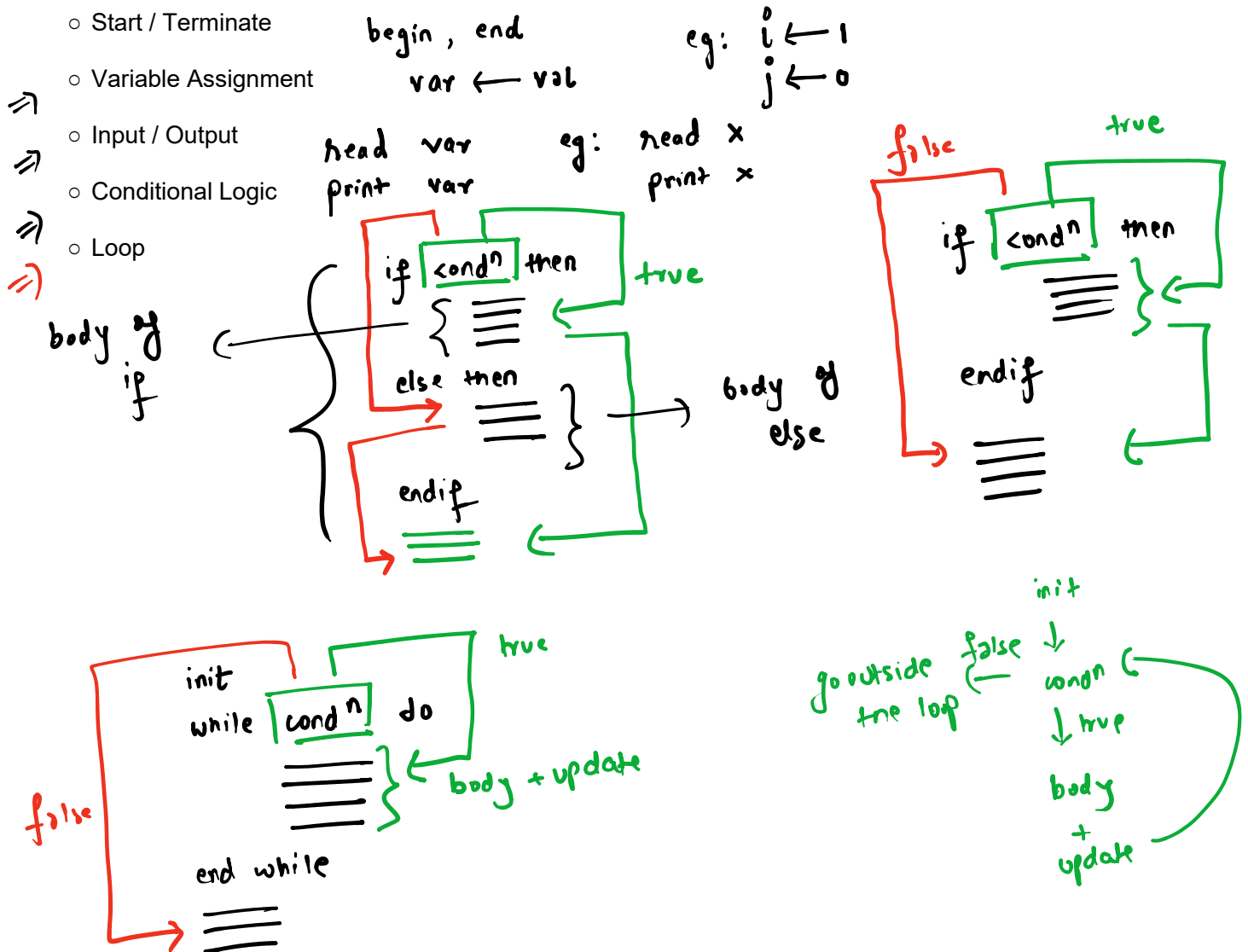


Pseudo-Code

It is a way of **expressing** an **algorithm** that involves usage of **plain** language to describe the steps of an algorithm.



Check Prime

Given a positive number $N (>1)$, design an algorithm to **check** if the given number is a **prime number** or **not** and express it by means of pseudo code.

Example

Input : N = 5
 Output : Prime

Input : N = 4
 Output : Not Prime

A number **N** is a **prime no.** if it is **divisible** by only **1** & the **number** itself.

$\boxed{5}$
n

$\boxed{2}$ ~~3~~ ~~4~~ 5
i

$\boxed{9}$ $\boxed{2}$ 3
n i

$2 \leq 8? \checkmark$
 $3 \leq 8? \checkmark$

$9 \div 2 = 0? \times$
 $9 \div 3 = 0? \checkmark$

```

begin
read n
i ← 2
while i ≤ n-1 do
    if  $n \div i == 0$  then
        ⇒ { print "not prime"
            end
        }
    endif
    i ← i+1
endwhile
print "prime"
end
    
```

$2 \leq 4? \checkmark$
 $3 \leq 4? \checkmark$
 $4 \leq 4? \checkmark$
 $5 \leq 4? \times$

$5 \div 2 = 0? \times$
 $5 \div 3 = 0? \times$
 $5 \div 4 = 0? \times$

```

begin
read n
i ← 2
while i ≤ n-1 do
    if  $n \div i == 0$  then
        print "not prime"
        end
    else then
        i ← i+1
    endif
endwhile
print "prime"
end
    
```

* Largest of N numbers

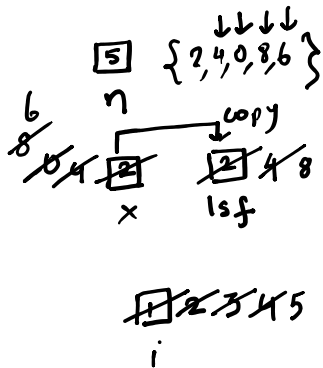
Given **N** integer values, design an algorithm to find the **largest** of the **N** numbers and express it by means of pseudo code.

Example

Input : N = 5

2	4	0	8	6
---	---	---	---	---

Output : 8 ↑ ↑ ↑



%p: 8

lsf = ~~2~~ ~~4~~ 8

```

begin
  read n
  head x
  lsf ← x
  i ← 1
  while i ≤ n-1 do
    {
      read x
      if x > lsf then
        lsf ← x
      endif
      i ← i+1
    }
  endwhile
  print lsf
end
  
```

body of while

body of if

```

begin
  read n
  lsf ← -∞
  i ← 1
  while i ≤ n do
    {
      read x
      if x > lsf then
        lsf ← x
      endif
      i ← i+1
    }
  endwhile
  print lsf
end
  
```

const work ← ls

INT-MIN (c++)

iterations = n

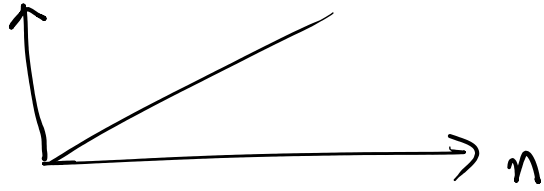
work done in each iteration = const

∴ total time

n · c

n(n)

end while
print lsf
end
time



~ $O(n)$
Linear time

Printing Stars

Printing Stars

Given a positive number **N**, design an algorithm to **print N stars** and express it by means of pseudo code.

Example

Input : N = 3

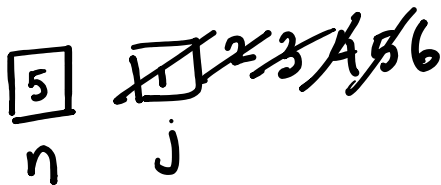
Output :

* * *

Input : N = 5

Output :

* * * * *



* * * * *



```
begin
read n
j ← 1
while  $j \leq n$  do
    print "*"
    j ← j + 1
endwhile
end
```

1 <= 5? ✓
2 <= 5? ✓
3 <= 5? ✓
4 <= 5? ✓
5 <= 5? ✓
6 <= 5? ✗

Matrix of Stars

Given a positive number N , design an algorithm to **print** a $N \times N$ **matrix of stars** and express it by means of pseudo code.

Example

 $N = 3$

```

* * *
* * *
* * *

```

 $N = 4$

```

* * * *
* * * *
* * * *
* * * *

```

begin
read n
 $i \leftarrow 1$
while $i \leq n$ do
 $j \leftarrow 1$
 while $j \leq n$ do
 print "*" body of inner-while loop
 $j \leftarrow j + 1$
 endwhile
 print "\n" body of outer while loop
 $i \leftarrow i + 1$
endwhile
end

time: $O(n^2)$
 $n=3$

```

* * *
* * *
* * *

```

```

* * *
* * *
* * *

```

print endl (slow)
or
print "\n" $\leftarrow$ newline char (fast)

~~1 2 3 4~~
↓

1 ≤ 3 ? ✓
2 ≤ 3 ? ✓
3 ≤ 3 ? ✓
4 ≤ 3 ? ✗

1 ≤ 3 ? ✓✓✓
2 ≤ 3 ? ✓✓✓
3 ≤ 3 ? ✓✓✓
4 ≤ 3 ? ✗

Case-1

```

|_|_|_|_|_|_|_|_|
|_|_|_|_|_|_|_|_|
|_|_|_|_|_|_|_|_|
|_|_|_|_|_|_|_|_|

```

track
platform
track
platform

Case 2

```

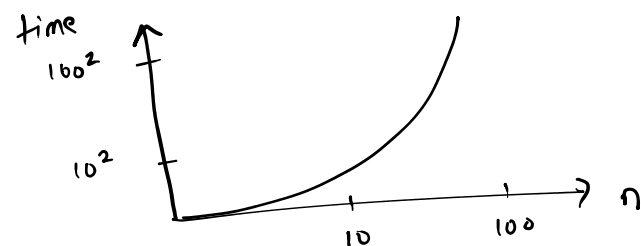
|_|_|_|_|_|_|_|_|
|_|_|_|_|_|_|_|_|
|_|_|_|_|_|_|_|_|
|_|_|_|_|_|_|_|_|

```

track
platform

} endl
newline
" \n "
escape char.

" \t "
tab



Right-Angled Triangle of Stars

Given a positive number **N**, design an algorithm to **print** a **right-angled triangle of stars** as shown in the example below and express it by means of pseudo code.

Example

N = 4

```
*
* *
* * *
* * * *
```

N = 5

```
*
* *
* * *
* * * *
* * * * *
```

N = 5

```
i=1 * . . . .
i=2 * * . . .
i=3 * * * . .
i=4 * * * * .
i=5 * * * * *
```

$1 \leq i \leq n$

#stars

```
1
2
3
4
5
i
```

n = 3

```

  *
 → * *
 → * * *
 → 3 4
  2
  *
  i

```

$1 \leq 3? \checkmark$
 $2 \leq 3? \checkmark$

begin

read n

i ← 1

while $i \leq n$ do

// for the ith row, print i stars

```

j ← 1
while j ≤ i do
  { print '*'
    j ← j + 1
  }
endwhile

```

```

1 2 3 4
j

```

$1 \leq 3? \checkmark$
 $2 \leq 3? \checkmark$

i
i

2 ≤ 3? ✓

3 ≤ 3? ✓

4 ≤ 3? ✗

endwhile
end

endwhile
print '\n'
i ← i + 1

2 ≤ 3? ✓
3 ≤ 3? ✓
4 ≤ 3? ✗

Right-Angled Triangle of Stars - 2

Given a positive number **N**, design an algorithm to **print** a **right-angled triangle of stars** as shown in the example below and express it by means of pseudo code.

Example

N = 4

```

      *
     * *
    * * *
   * * * *
  
```

N = 5

```

      *
     * *
    * * *
   * * * *
  * * * * *
  
```

N = 5

```

i=1  - - - - *
i=2  - - - * *
i=3  - - * * *
i=4  - * * * *
i=5  * * * * *
1 ≤ i ≤ n
  
```

	#spaces	#stars
i=1	4	1
	3	2
	2	3
	1	4
	0	5
	<u>1-i</u>	<u>i</u>

begin

read n

i ← 1

while i ≤ n do

// for the ith row, print n-i spaces

// for the i^{th} row, print $n-i$ spaces

$j \leftarrow 1$

while $j \leq n-i$ do

print ' '

$j \leftarrow j+1$

endwhile

// followed by i stars

$k \leftarrow 1$

while $k \leq i$ do

print '*'

$k \leftarrow k+1$

endwhile

print '\n'

$i \leftarrow i+1$

endwhile

end

Pyramid of Stars

Given a positive number **N**, design an algorithm to **print** a **pyramid of stars** as shown in the e.g. below & express it by means of pseudo code.

Example

N = 5

$N = 5$ last quer.

A diagram showing a 5x5 grid of asterisks. A vertical blue line is drawn between the third and fourth columns. Above the line, there is a blue arrow pointing left and a blue arrow pointing right, both originating from a central point on the line. This represents a split or partitioning of the grid.

N = 5

[illegible]

begin

head n

 $i \leftarrow 1$

```
while i <= n do
```

// print n-i spaces

$$j \leftarrow 1$$

```
j ← 1
while j ≤ n-i do
  print ' '
  j ← j+1
```

```

    print
    j ← j+1
endwhile
// print i stars
j ← 1
while j ≤ i do
    print '*'
    j ← j+1
endwhile
// print i-1 stars
j ← 1
while j ≤ i-1 do
    print '*'
    j ← j+1
endwhile
i ← i+1
print '\n'

```

end while
end

```

i=1 - - - - *
i=2 - - - * * *
i=3 - - * * * * *
i=4 - * * * * * * *
i=5 * * * * * * * * *

```

$1 \leq i \leq n$

#spaces	#stars
4	1
3	3
2	5
1	7
0	9
<u>$n-i$</u>	<u>$2i-1$</u>

begin
read n

```

i ← 1
while i ≤ n do
    // print n-i spaces
    j ← 1
    while j ≤ n-i do
        print ' '
        j ← j+1
    endwhile
    // print 2i-1 stars
    j ← 1
    while j ≤ 2i-1 do
        print '*'
        j ← j+1
    endwhile

    i ← i+1
    print '\n'
endwhile
end

```

Homework

Try to write the pseudo code for the following patterns

